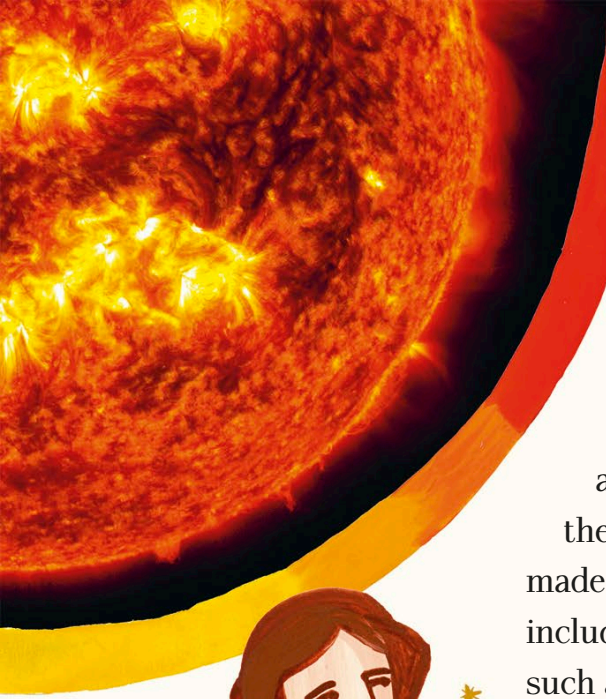




Cecilia explained how the mixture of different colors of light given out by a star tells us the temperature at its fiery surface.

Hydrogen and helium

For years, scientists had looked at the spectrum of light from the sun and decided that it was made of the same stuff as the Earth—including mostly heavy elements, such as iron and calcium. Other stars had different spectra, so scientists assumed they must be made of different elements. Cecilia realized this was wrong.

Cecilia discovered that the differences between light spectra from different stars were not due to different ingredients, but different temperatures. She also came up with a way to calculate how much of each element was present in a spectrum. Cecilia was surprised to discover that the sun is mainly made of hydrogen and a little bit of helium. She shared her breakthrough with other leading astronomers. At first, they said that this was impossible. But, gradually, Cecilia was proven right—our understanding of stars and the universe had changed forever.

This missing color has been absorbed by oxygen.



Cecilia had to battle for recognition, but eventually became Harvard's first female professor and head of the astronomy department.

Solar spectrum

Sunlight looks white, but it is really a mixture of many colors. A spectrum looks a little like a colorful barcode, where each dark line shows us a specific color of light that is missing. These colors are missing because they have been absorbed by elements in the star's atmosphere. If we know which colors of light are absorbed by which elements, we can figure out what elements make up a specific star.

